**Group No: 1**

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**Title:** Aquarium Water Quality Control System
**Date:** March 1, 2025

## Full Description

This project focuses on an **automated aquarium water quality control system** that maintains optimal conditions for aquatic life. The system integrates multiple sensors and actuators to regulate **temperature, turbidity, and pH levels** efficiently.

1. **Temperature Control**
    - A **DS18B20 waterproof temperature sensor** continuously monitors the water temperature.
    - When the temperature drops below the desired threshold, a **submersible aquarium heater(25W)** is activated via a **relay module**.
    - The Arduino microcontroller ensures temperature stability by toggling the heater **ON/OFF** based on real-time readings.
2. **Turbidity Control**
    - A **SKU SEN0189 turbidity sensor** detects water clarity levels.
    - When turbidity exceeds the set limit, a **DC12V water pump** is activated to initiate filtration and improve water clarity.
3. **pH Stabilization**
    - A **PH-4502C Liquid pH Sensor** measures the water's pH level.
    - If the pH deviates from the optimal range, a **G328 peristaltic pump** adds a stabilizing solution (acid or base) to correct the pH.
    - The **relay module** controls the peristaltic pump to ensure accurate dosing.

## Main Components

### Microcontroller

- **ESP32 –** Handles sensor readings, actuator control, and Wi-Fi connectivity for remote monitoring.

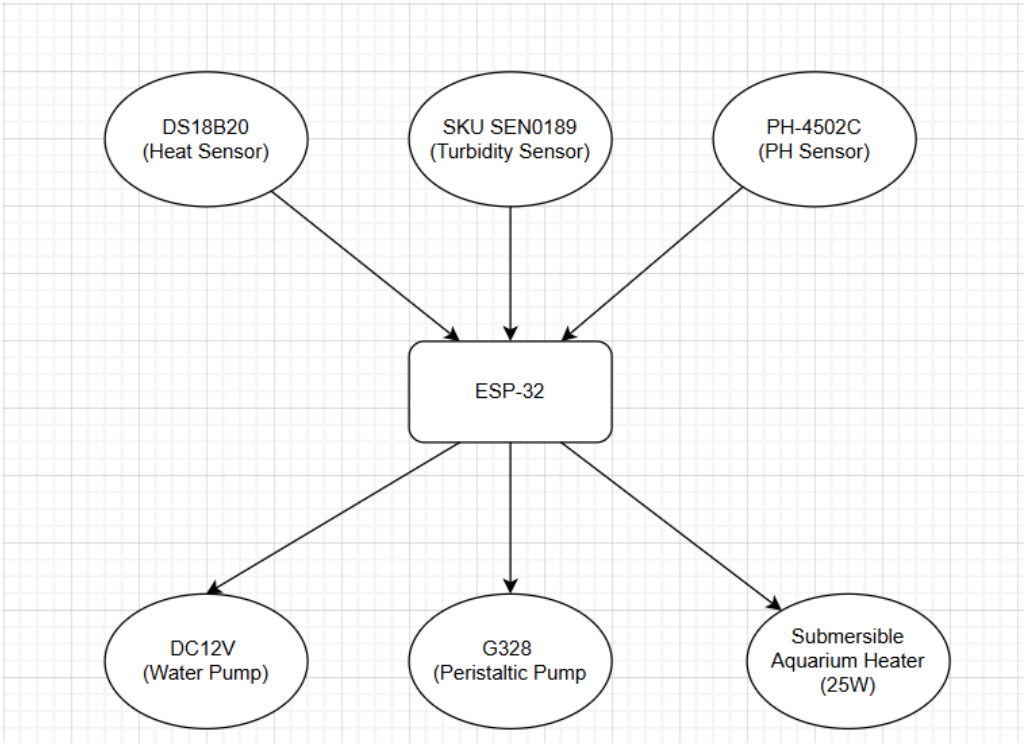
### Inputs (Sensors)

- **DS18B20** – Waterproof Temperature Sensor
- **SKU SEN0189** – Turbidity Sensor
- **PH-4502C Liquid pH Sensor** – pH Monitoring

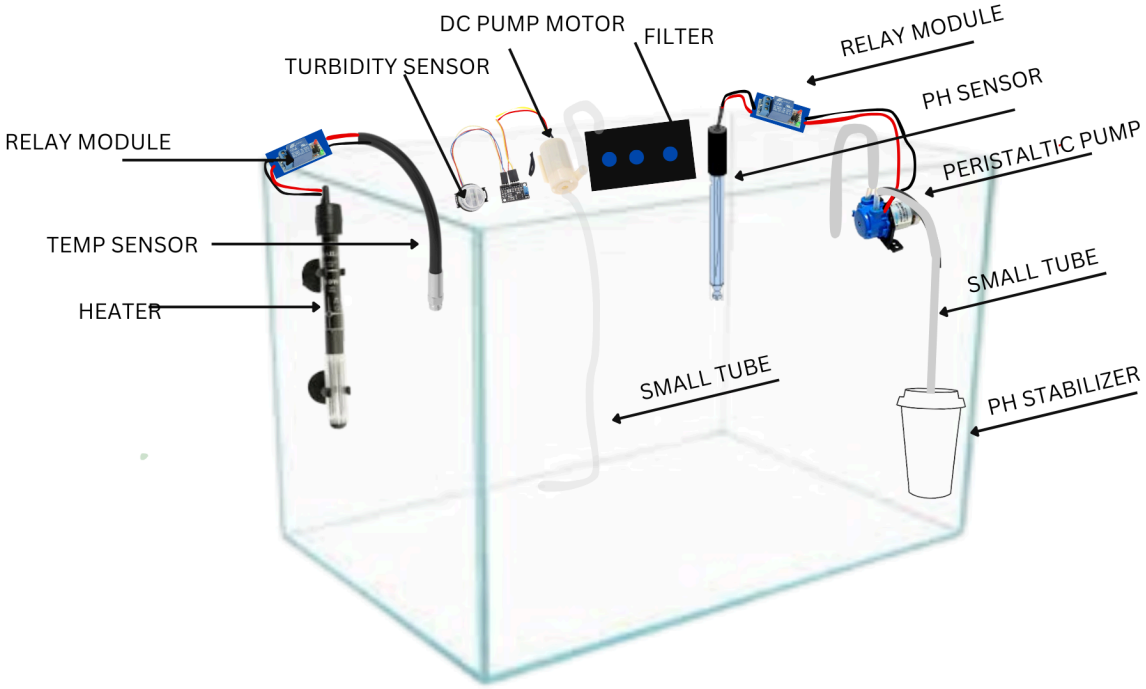
### Outputs

- **DC12V Water Pump** – For turbidity control
- **G328 Peristaltic Pump** – For pH adjustment
- **Submersible Aquarium Heater (25W)**– For temperature regulation
- **SRD-05VDC-SL-C Relay Module** – Controls heater and peristaltic pump

## Block Diagram



## Expected Output (Digitalized Drawing)




**Reference links**

Controlling a Relay using Temperature (Temperature Controlled Relay with Thermocouple, Arduino, LCD)

How to Make an Amazing DIY Top Filter for Your Aquarium at Home!

pH Sensor and Peristaltic Pumps

How to use & Calibration pH Meter DFRobot Water quality testing With Arduino